

Turn Your Rooftop and Backyard Into **PV Solar Plants**

Very soon, the government will be buying power generated from rooftop and backyard solar power plants. Notwithstanding what it decides, it is important to look at the key considerations in evaluating solar option for your rooftop and have a basic understanding of the components of a solar power system and how it generates electricity



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Solar energy is a hot topic today. Some of the factors to consider before installing a solar power plant on the rooftop of your building include electrical load, current rate, roof size, load capacity and geographic location of the building. The subsidy given by the government, local regulations and incentives are also key determinants in evaluation.

Rooftop solar arrays are best installed on a large, flat roof where direct sunlight is available without shadows from the surrounding structures. If there is shadow on a part of the terrace during the day, photovoltaic solar panels will not be able to harvest the sun's energy for that period.

Rooftop and backyard installations: Growth drivers

In the USA, the total number of grid-tied rooftop photovoltaic (PV) solar power projects grew by 81 per cent in 2008 (to 196 MW) and an additional 93 per cent in 2009 (to 385 MW). Non-residential rooftop and backyard installation was the fastest-growing solar market that grew at 109 per cent. This segment includes rooftop PV solar installations for commercial and government buildings, factories, warehouses, schools, colleges and universities. In fact, solar power projects have been growing by over 62 per cent a year for the past five years in the US.

You may ask why is solar power growing the world over. Are factors beyond climate change concerns driv-

ing the growth? Many of us believe that this will happen in India too. There are great merits in promoting installation of solar plants especially on rooftops and backyards lying unused today. In fact, India should promote them in preference to megawatt plants.

Let us consider how many thousand square kilometres of rooftops and backyards in India are currently lying unused. Like in USA, India too has commercial and government buildings, factories, warehouses, schools, colleges and universities with very large unused terraces and backyards. We can have at least a gigawatt of solar power pushed into the grid to augment shortages. With small rooftop solar power plants feeding into the copper grid at far end of the grid, quality of the power to every other customer in the vicinity will improve during the productive hours in daytime. Terminal points of the grid suffer from high source impedance and solar power injection will help to lower it and consequently stabilise the line voltage.

Say, 5MW solar energy from a hundred 50kW rooftop or backyard solar power plants is fed into a small city grid like Miraj, Ratnagiri or Ratlam. All consumers in that area will get cleaner uninterrupted power from the local copper grid. The high impedance of the local grid helps power to remain local and thus instantly helps improve local power quality. That's why all other countries in the world allow solar plant owners to feed power into the local grid at low voltage end.

Many would not know that total